GEOTRACES Intercalibration Report

Cruise ID*: **RR1815**
Submitting investigator*: **Alan Shiller - University of Southern Mississippi - alan.shiller@usm.edu**
Parameters to be intercalibrated*:

- Tb_D_CONC_BOTTLE::whliao pmol/kg
- Sm_D_CONC_BOTTLE::4ekw3t pmol/kg
- Pr_D_CONC_BOTTLE::kytybd pmol/kg
- Er_D_CONC_BOTTLE::dxdypm pmol/kg
- Eu_D_CONC_FISH::60umzp pmol/kg
- Er_D_CONC_FISH::flv4pg pmol/kg
- Ho_D_CONC_BOTTLE::qafdfm pmol/kg
- Y_D_CONC_FISH::asoboe pmol/kg
- Lu_D_CONC_BOTTLE::dgzysp pmol/kg
- Yb_D_CONC_BOTTLE::aebv5r pmol/kg
- Ce_D_CONC_BOTTLE::7r1crl pmol/kg
- Tm_D_CONC_FISH::vshlpq pmol/kg
- Y_D_CONC_BOTTLE::vbtqm7 pmol/kg
- Gd_D_CONC_FISH::vv3b0x pmol/kg
- Dy_D_CONC_FISH::drvjhn pmol/kg
- Lu_D_CONC_FISH::jcdlov pmol/kg
- Sm_D_CONC_FISH::3j9pqw pmol/kg
- Eu_D_CONC_BOTTLE::vbw6j8 pmol/kg
- Ce_D_CONC_FISH::nrzhse pmol/kg
- Tm_D_CONC_BOTTLE::d8qf2f pmol/kg
- Gd_D_CONC_BOTTLE::fgu3vr pmol/kg
- Tb_D_CONC_FISH::j519ko pmol/kg
- Yb_D_CONC_FISH::7thabf pmol/kg
- La_D_CONC_BOTTLE::19sfbu pmol/kg
- La_D_CONC_FISH::l1u3qy pmol/kg
- Dy_D_CONC_BOTTLE::qhjczn pmol/kg
- Ho_D_CONC_FISH::gjxtdt pmol/kg
- Pr_D_CONC_FISH::xp0hyb pmol/kg
- Nd_D_CONC_FISH::ea16lv pmol/kg
- Nd_D_CONC_BOTTLE::4vjjtn pmol/kg

Note: Nd data were originally submitted to IDP21 but are included here both because they are part of the REE series and also because we are submitting revised data (a few samples were incorrectly listed in the 2021 submission).

***Once generated, these headings must not be changed or altered.**

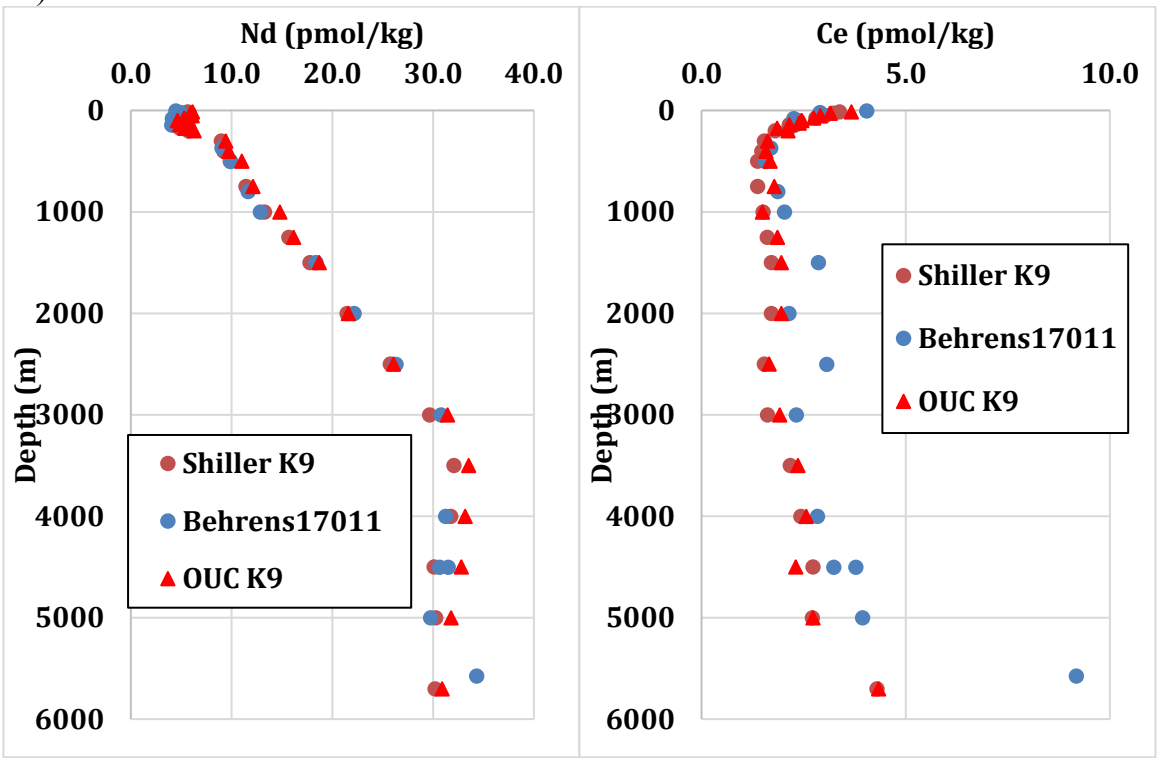
Preamble: The responses provided here will be evaluated by 6-8 members of the GEOTRACES intercalibration committee to verify that the data have been intercalibrated prior to their dissemination in the upcoming IDP.

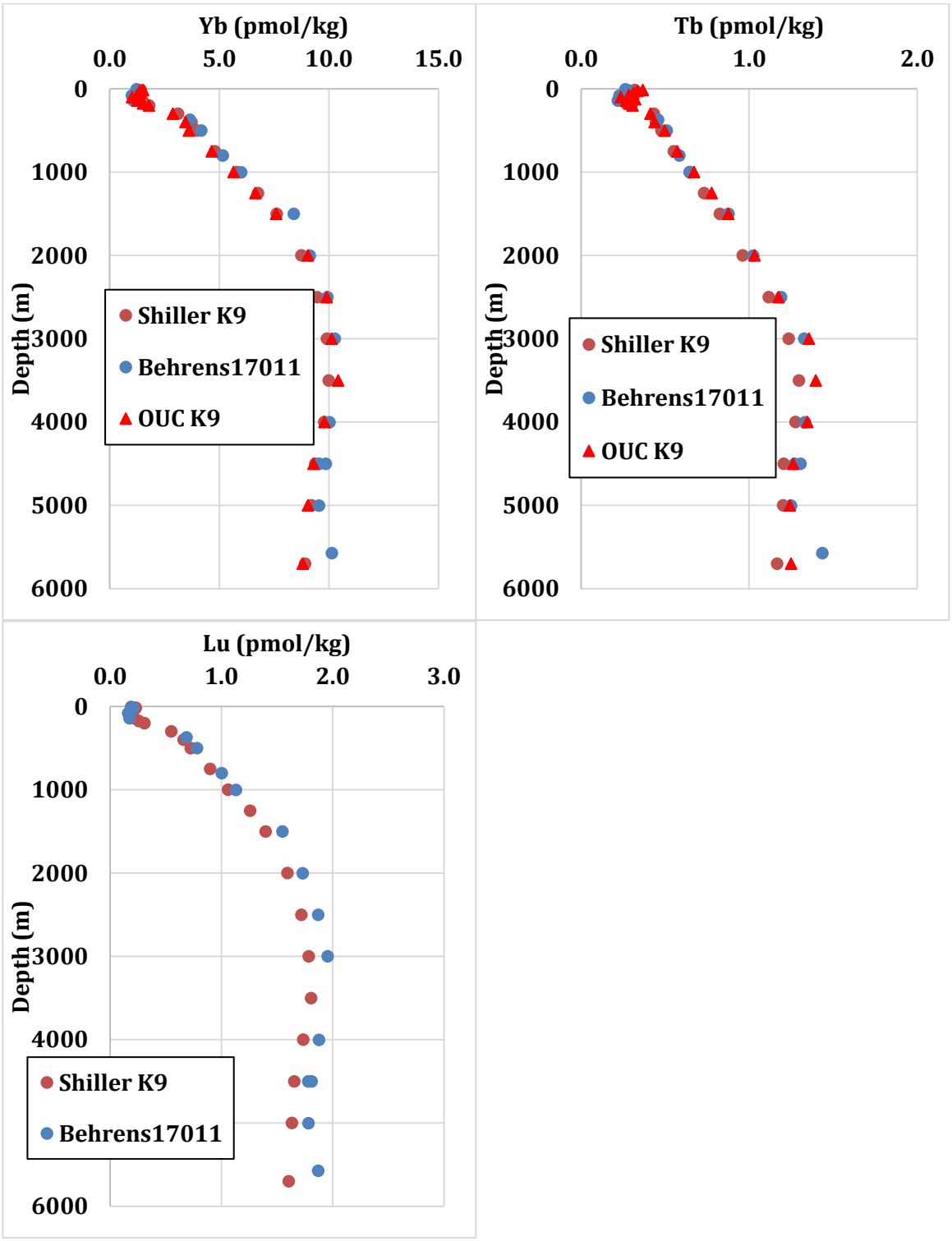
Please fill in as many sections as possible. For reference, an example of intercalibration report, dully filled in, is

provided **here**. You can also **download an example** of report for nutrient data. Note that missing information may delay approval. Please contact us with questions (**intercal@geotraces.org**).

1. **If your lab participated in an intercalibration exercise (**http://www.geotraces.org/sic/intercalibrate-data/intercalibration-exercices**), please**
   a) provide a relevant figure or table,
   b) describe the results, identifying your laboratory,
   c) provide a reference, if published.

   For the rare earth elements, we participated in a GP09 intercalibration with Jing Zhang's group at the Ocean University of China. Note that those samples were analyzed contemporaneously with our GP15/RR1814/RR1815 samples. For most of the series, the comparison is excellent and is summarized in Cao et al., 2024, https://doi.org/10.1016/j.gca.2024.02.018. Below we provide representative GP09 comparison figures showing dissolved REE profiles: USM results vs OUC results for the same profile (Station K9; 11N, 149E). Also shown for comparison are nearby results from Behrens et al. Geochim Cosmochim Acta 222 [2018] 508, Station GeoB17011 at 10.6N, 149E. Note that OUC did not report Lu data. Overall, the comparison is quite good. (Note that REE data at 18N, 158E from Deng et al. [Sci. Rept. 7, 2019, 16539] is significantly higher than the data shown in the graphs below. For example, they show Nd = 27 pM at 1000 m and Ce >15 pM at depths >1000 m.)





2. **Please describe your sampling methodology at sea. Highlight how it followed or deviated from the GEOTRACES cookbook (available at: http://www.geotraces.org/cookbook)? (sampling apparatus, cleaning procedures, type and size of filters, sample processing and storage…).**

Yes, we followed the GEOTRACES cookbook. Clean seawater samples were collected using a GEOTRACES CTD referred to as GT-C/12L GoFlo, and also from the Super-GeoFISH towed surface vehicle. For more information, see the cruise report.

Water samples were filtered through pre-cleaned, 0.2 µm Pall Acropak Supor filter capsules as

described elsewhere (e.g., Cutter et al., 2014; Hatta et al., 2015). Filtered water was collected in 125 mL HDPE bottles (Nalgene) that had been precleaned by soaking in hot 1.2 M HCl (reagent grade) for at least 8 h with subsequent thorough rinsing with ultrapure distilled deionized water (Barnstead E-pure).

3. Briefly outline the analytical methodology used in your laboratory, and provide associated metadata and references, as appropriate.

Dissolved Rare Earth Elements (REEs including Y), were determined using 14 mL of sample which-was spiked with a mixture of isotopically-enriched Nd-145, Sm-149, Eu-153, Gd-155, Dy-161, Er-167, and Yb-171 (Oak Ridge Nat'l. Labs). Each spike was >90% enriched in the listed isotopes. The sample/spike ratio was chosen so as to have the analytical isotope ratios approximately the geometric mean of the natural and enriched spike isotope ratios. Samples were then extracted/pre-concentrated using a SeaFAST system (Elemental Scientific, Inc.) operated in offline mode. A similar online SeaFAST extraction procedure is described by Hathorne et al., 2012. The extracted samples were subsequently analyzed using a Thermo-Fisher high resolution ICP-MS with an Apex-FAST high efficiency sample introduction system with Spiro desolvator (Elemental Scientific, Inc.). The instrument was operated in low resolution. The enriched isotope spikes also served to provide counts/sec. calibration factors for elements that were not spiked with enriched isotopes. This calibration was also examined with a standard made in dilute nitric acid. Precision and recovery were checked by analysis of a large-volume composite seawater sample. Spiked (with a natural isotopic abundance elemental spike) and unspiked aliquots of this sample were analyzed twice in each analytical run. A Ba standard was also run to check for BaO^+ interference on several isotopes and Ba in the extracted samples was also monitored. Because the extraction resin in the SeaFAST system (Nobias PA-1) discriminates against Ba, plus the reduction of the BaO^+ interference by the desolvation system, BaO+ was less than 0.1% of the counts in Eu-151, Eu-153, Gd-155, and Gd-157. Tests also revealed no significant low REE oxide interference on mid-/high-REEs.

The reproducibility error of this method was estimated by comparing samples collected at the same depths on different casts at the same station (GA03 and GP15) as well as by repeated measurement of GEOTRACES reference waters and an in-house standard. Recovery of the method was determined by repeated analysis of a spiked and unspiked seawater. The recoveries, precisions, and comparisons to reference waters are shown in Tables 1-3 for the dissolved concentration data.

4. Explain how blank values and detection limits were defined and evaluated. Report if possible in a table format:
   a) blank values
   b) detection limit
   c) relative contribution to sample values (if applicable to your parameter)

Blank and detection limit values are shown in Table 1. In general, the blank is evaluated by measurements of: a) ultrapure HCl used for acidifying samples (usually negligible), b) reagents used in SeaFAST extractions, c) ultrapure HNO3 or H2O used for sample dilution, and d) air blank of SeaFAST system. Detection limits are determined by 3x the standard deviation of the blank or of a low-analyte sample.

5. Explain how you monitored the internal consistency of your data (e.g. number and type of replicates). Report the results (e.g. rsd), if possible in a table format.

Internal consistency is monitored several ways. Mainly we use several large volume in-house seawater samples (usually consisting of 'excess' cleanly collected sample waters obtained during different GEOTRACES cruises). These samples are large enough so that we can analyze them for multiple (often >30) analytical runs spanning more than one cruise. They are generally analyzed twice during each analytical run. This allows us to compare reproducibility, standardization and drift over several years. Usually, one large volume sample is a spiked version of another sample, thus allowing us to estimate recovery. Where possible, we will also compare cast overlap samples as well as GEOTRACES reference waters. In general, we do not run replicate analyses due to the large number of samples per cruise and need to run several analytical methods on each sample. Odd results (e.g., sample results that don't fit typical profiles) are reanalyzed along with a small number of randomly selected 'good' samples. Consistency data is provided in Table 2.

6. **Explain how you evaluated the external consistency of your data (e.g., results from analyses of certified reference materials and/or consensus materials). Report the results, if possible, in a table format.**

Table 3 includes data on GEOTRACES reference waters and literature comparisons are shown in the figures below.

7. **If you occupied a crossover station (please see intercalibration procedures: https://www.geotraces.org/intercalibration-procedures/) include a plot and a table that show relevant data and their level of agreement, and explain any significant discrepancies by providing supporting information (e.g. profile of temperature & salinity, potential density).**

For Leg 1 of GP15 (RR1814), there was a crossover station between GP15-Stn 8 and the 2017 Japanese GP02-Stn CL09 (also called KH17-03 or GP02-bis). At our last inquiry, REE data were not available for GP02.

For Leg 2 of GP15 (RR1815), there is a crossover between GP15-Stn 35 and GP16-Stn 36. Both are US cruises using the same sampling equipment. However, the group doing REEs on GP16 did not analyze Stn 36, the closest station they analyzed was Stn 34.

We show graphs of these GP15/GP16 comparisons below. In general, these comparisons show that the two labs produce similar data, with the exception of Ce.

8. **If available, report replicate analyses from a different laboratory, and/or explain how your data compare to historical data including results from nearby stations, even though they may not be true crossover stations.**

We provide comparison graphs for Nd, Yb, and Ce as well as for Y. Some commentary regarding Ce data is also provided with sections shown below.

9. **If you could not provide answer to either questions 7 or 8, please provide an explanation (e.g., state-of-the-art, first time measurement of a given parameter)**

10. **If not already included in your responses to the questions above, please provide a representative vertical profile or report the range of values, for the parameter(s) that are addressed in this intercalibration report.**

Once completed, please upload the report here:
https://geotraces-portal.sedoo.fr/pi/

Thank you! If you need any clarification, don't hesitate to contact the S&I committee: intercal@geotraces.org

See Tables and Figures on following pages.

Table 1: Blanks, detection limits, recoveries, and range of values for rare earth elements.

	Y pmol/kg	La pmol/kg	Ce pmol/kg	Pr pmol/kg	Nd pmol/kg	Sm pmol/kg	Eu pmol/kg	Gd pmol/kg	Tb pmol/kg	Dy pmol/kg	Ho pmol/kg	Er pmol/kg	Tm pmol/kg	Yb pmol/kg	Lu pmol/kg
Typical Blanks	0.12	0.09	0.11	0.01	0.09	0.04	0.03	0.021	0.006	0.079	0.001	0.010	0.004	0.006	0.004
Typical detection limits (Note 1)	0.37	0.26	0.34	0.04	0.26	0.11	0.10	0.06	0.019	0.24	0.004	0.029	0.013	0.017	0.011
Low sample based detn limit (Note 2)	8.06	0.71	0.49	0.10	0.57	0.14	0.07	0.22	0.048	0.28	0.062	0.237	0.045	0.208	0.036
Spike Recovery (Note 3)															
For GP15															
Percent recovery	108.5 %	100.8 %	99.1 %	97.7 %	98.5 %	97.9 %	99.0 %	95.0 %	97.0 %	97.6 %	94.7 %	98.3 %	92.6 %	96.4 %	90.0 %
RSD	6.3%	6.1%	5.5%	5.2%	4.6%	6.1%	6.3%	5.7%	6.3%	5.6%	5.2%	8.1%	8.0%	4.7%	4.2%
Replicates	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19
Range of Concentrations															
Minimum	53	2.59	0.81	0.53	2.58	0.49	0.15	0.95	0.15	1.21	0.40	1.18	0.15	0.62	0.10
Maximum	418	99	7.7	17.1	74.6	14.6	3.9	18.6	3.0	20.5	5.1	15.2	2.4	17.6	3.0

NOTES

1. Detection limits based on acid and extraction blanks. Detection limit will vary with sample pre-concentration/dilution.
2. Detection limit estimated from uncertainty in repeated analyses of a low analyte sample.
3. Spike recovery shows percentage recovery based on the difference between a metal-spiked and unspiked seawater sample.

Table 2: Internal consistancy metrics based on in-house reference waters for rare earth elements.

	Y pmol/kg	La pmol/kg	Ce pmol/kg	Pr pmol/kg	Nd pmol/kg	Sm pmol/kg	Eu pmol/kg	Gd pmol/kg	Tb pmol/kg	Dy pmol/kg	Ho pmol/kg	Er pmol/kg	Tm pmol/kg	Yb pmol/kg	Lu pmol/kg
"NAZTJ"	n=24	n=24	n=24	n=24	n=24	n=24	n=24	n=24	n=24	n=24	n=24	n=24	n=24	n=24	n=24
Concentration	134.8	22.38	6.50	3.12	13.85	2.50	0.68	3.85	0.59	4.55	1.25	4.24	0.62	3.95	0.67
Std. deviation	5.8	0.54	0.19	0.04	0.18	0.06	0.02	0.09	0.01	0.12	0.02	0.06	0.06	0.05	0.01
RSD	4.3%	2.4%	2.8%	1.4%	1.3%	2.4%	2.3%	2.4%	1.6%	2.7%	1.4%	1.5%	9.1%	1.2%	1.6%
"NAZTJ-Spike"															
Concentration	488.6	108.9	20.26	19.79	90.04	16.63	6.07	21.81	4.80	25.91	6.68	23.74	3.53	22.32	3.90
Std. deviation	35.5	11.18	0.49	0.27	1.42	0.27	0.10	0.35	0.09	0.65	0.09	0.41	0.08	0.28	0.08
RSD	7.3%	10.3%	2.4%	1.4%	1.6%	1.6%	1.7%	1.6%	1.8%	2.5%	1.4%	1.7%	2.3%	1.3%	2.0%

GP13455 for GP09, GP15, and GP17-OCE															
Replicates	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19
Concentration	342.8	72.37	4.34	12.13	52.03	10.16	2.72	13.6	2.06	14.2	3.58	11.6	1.72	12.1	2.04
Std. deviation	22.53	2.64	0.20	0.47	2.03	0.42	0.11	0.69	0.06	0.52	0.09	0.45	0.06	0.43	0.08
RSD	6.6%	3.7%	4.6%	3.9%	3.9%	4.1%	4.0%	5.0%	3.0%	3.6%	2.6%	3.9%	3.4%	3.5%	4.0%

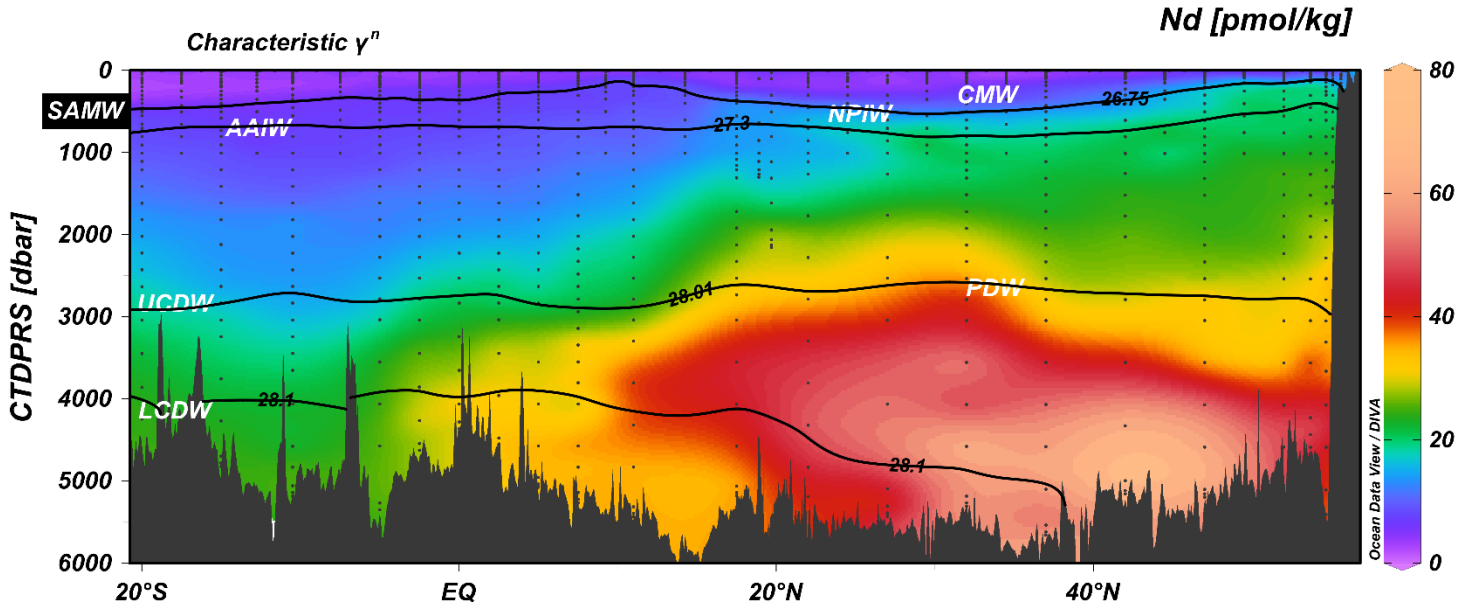
GP13609 for GP09, GP15, and GP17-OCE															
Replicates	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19	n=19
Concentration	77.7	8.35	3.40	1.69	7.59	1.63	0.43	2.29	0.34	2.29	0.59	1.87	0.23	1.30	0.20
Std. deviation	6.5	0.42	0.19	0.05	0.26	0.06	0.02	0.10	0.01	0.08	0.02	0.06	0.01	0.07	0.01
RSD	8.3%	5.0%	5.7%	2.8%	3.4%	3.6%	4.4%	4.5%	4.2%	3.7%	3.2%	3.3%	3.4%	5.1%	3.1%

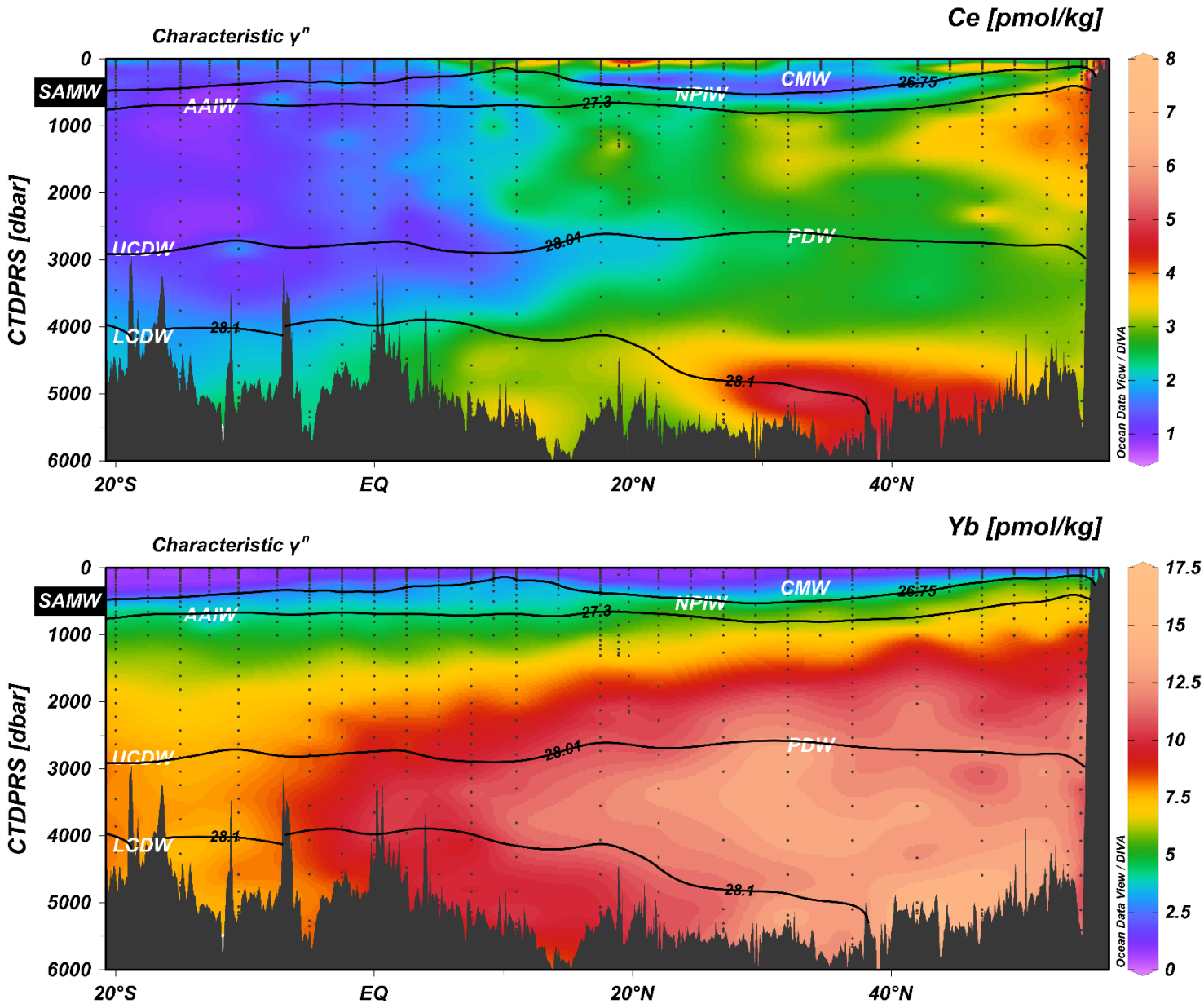
Table 3. Results for US GEOTRACES Reference Waters for rare earth elements.

Concentrations in pmol/kg

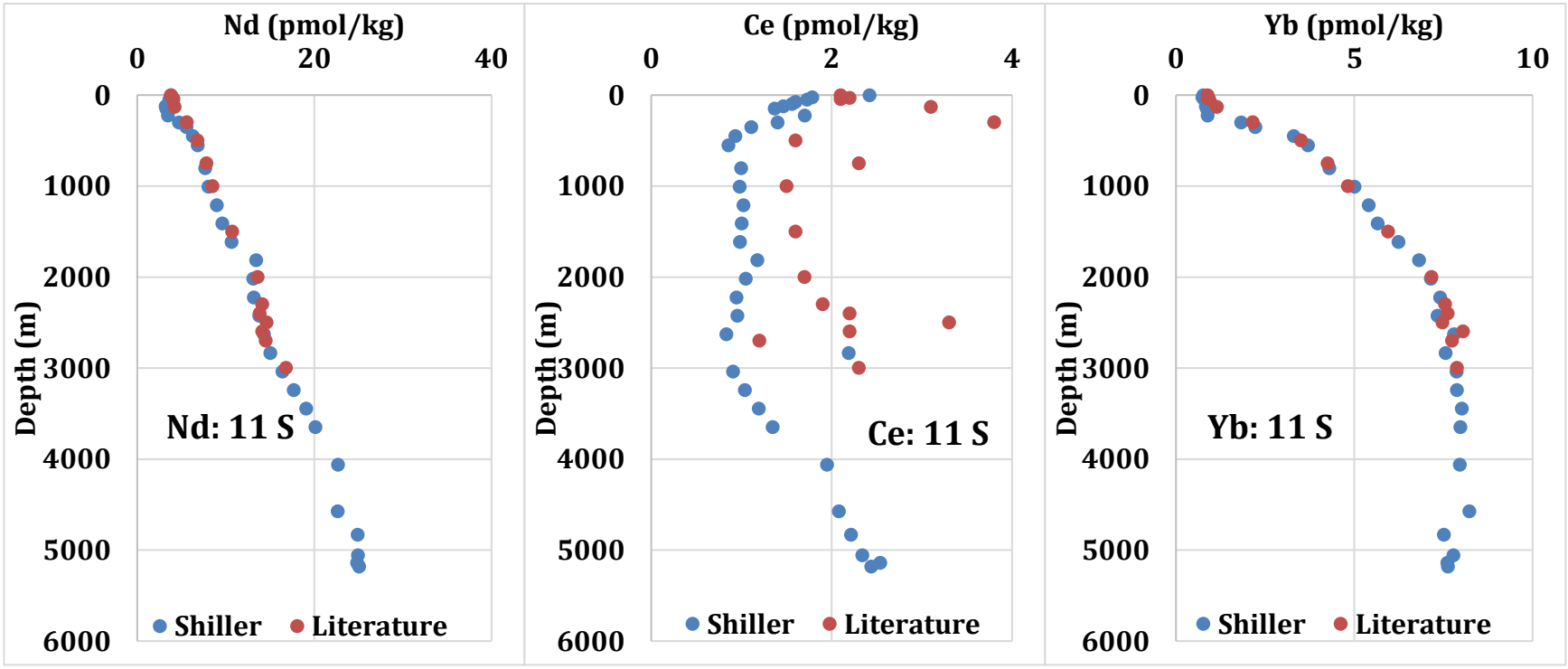
	Y	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
GSC-93	129.1	20.6	16.1	3.35	14.12	2.58	0.667	3.72	0.560	4.11	1.09	3.57	0.514	2.99	0.518
RSD (n=5)	4.8%	3.8%	4.1%	3.3%	1.9%	2.1%	2.7%	1.8%	2.3%	0.9%	1.8%	1.2%	4.8%	2.2%	5.1%
GSP-29	71.3	5.73	3.06	0.887	3.92	0.758	0.218	1.33	0.214	1.60	0.452	1.45	0.184	0.90	0.143
RSD (n=4)	2.9%	5.7%	2.8%	5.2%	3.6%	3.7%	3.3%	4.7%	3.9%	4.0%	0.7%	3.8%	2.0%	5.3%	5.8%

In the three sections below, we show the distributions of Nd, Ce, and Yb for our GP15 data. These sections indicate that the data show broad consistency across the section for all three elements, with minimal patchiness (which often suggests scattered data indicative of analytical artefacts and/or contamination). Below the sections we show profile comparisons with published data from the literature. In general, our Nd and Yb data compare well with published data. The one REE for which the comparisons are not very good is Ce. Because our Ce section shows broad meridional consistency (as opposed to the often scattered data of other published profiles), we believe our data our reasonable. One likely reason for the scattered nature of many published Ce profiles is that REEs are often viewed as not being significantly affected by contamination. Yet, not only does Ce have various industrial applications, but it is also the most concentrated REE in the crust (~66 ppm) but is generally less soluble than other REEs due to its oxidation to the particle-reactive +IV oxidation state. Many published REE measurements have been done on samples taken from 'regular' Niskin-type bottles as opposed to trace element clean bottles such as acid-washed, teflon-coated Go-Flos or Niskin-X bottles, which may contribute to contamination issues with Ce. We note that the DeBaar et al. data, with which we have reasonable agreement, are from samples collected using clean, Teflon-coated Go-Flos.

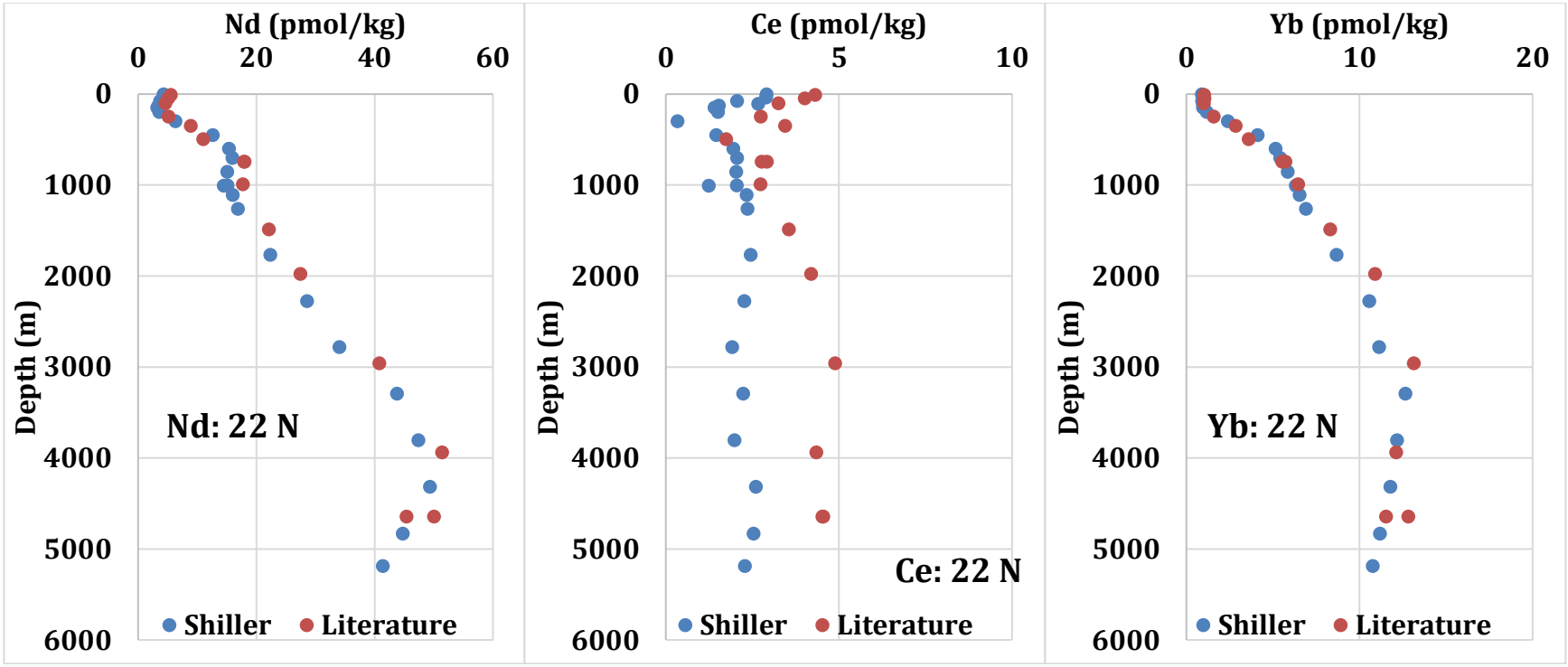



Ce [pmol/kg]
Characteristic γ^n
CTDPRS [dbar]
SAMW
AAIW
27.3
NPIW
CMW
26.75
UCDW
28.04
PDW
LCDW
28.1
28.1
0
1000
2000
3000
4000
5000
6000
20°S
EQ
20°N
40°N
8
7
6
5
4
3
2
1
Ocean Data View / DIVA
Yb [pmol/kg]
Characteristic γ^n
CTDPRS [dbar]
SAMW
AAIW
27.3
NPIW
CMW
26.75
UCDW
28.04
PDW
LCDW
28.1
28.1
0
1000
2000
3000
4000
5000
6000
20°S
EQ
20°N
40°N
17.5
15
12.5
10
7.5
5
2.5
0
Ocean Data View / DIVA

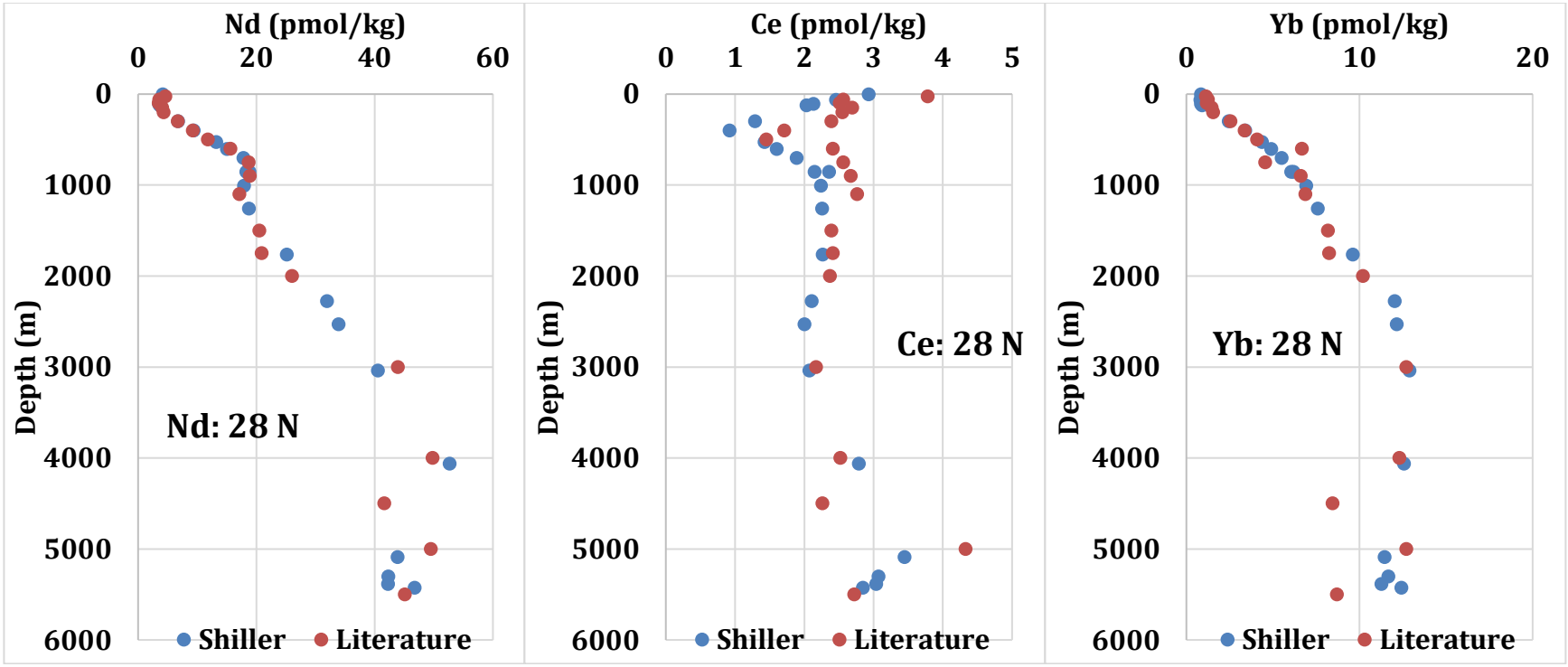
GP15-Station 35; 10.5S, 152W (Shiller) with GP16-Station 34 (Haley et al., DOI:10.26008/1912/bco-dmo.915447.1 and Haley et al., Earth Planet. Sci. Lett. 2021, https://doi.org/10.1016/j.epsl.2021.117233). Note that Haley et al. did not report data for the GP16 crossover station, but rather another nearby station. Comparisons are good, except for Ce where their data are high and scattered.



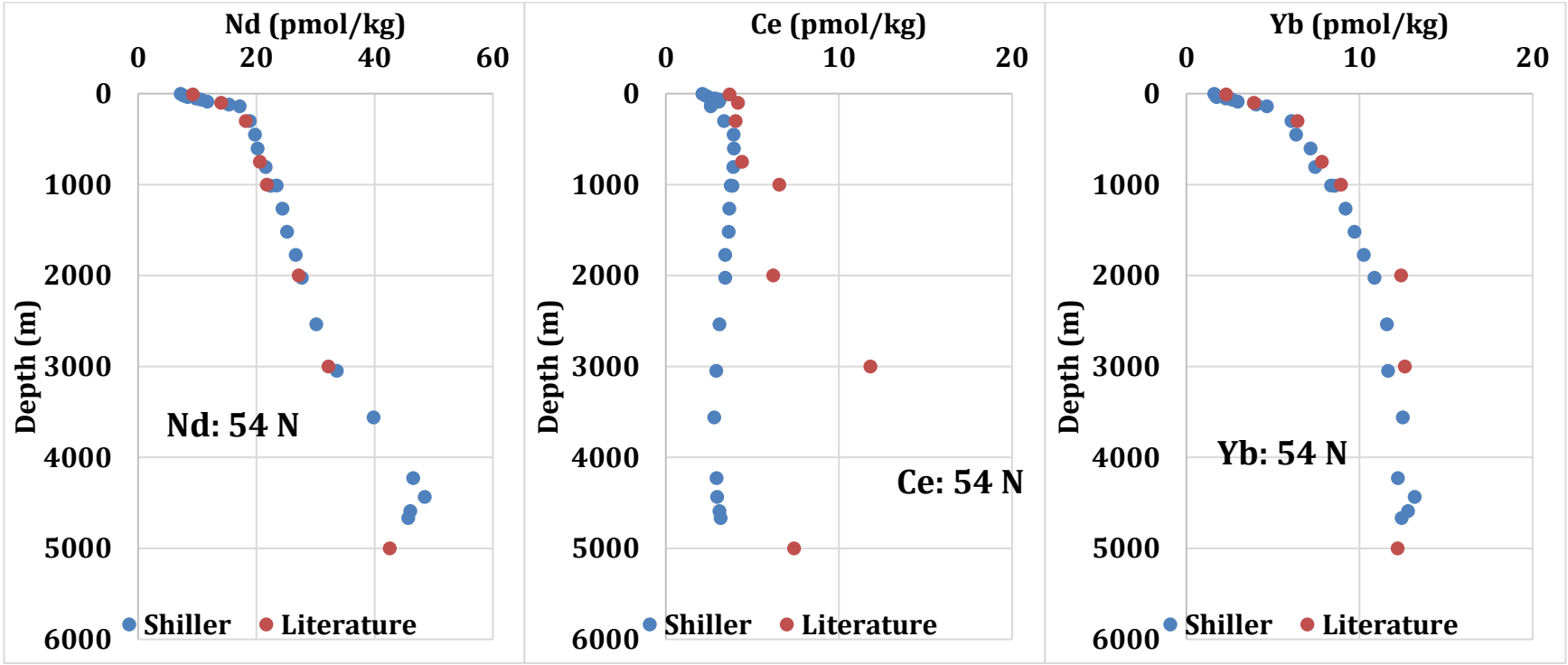
GP15-Station 18; 22N, 152W (Shiller) and Station Aloha 22.75N, 158W (Frollje et al., Geochim. Cosmochim. Acta 2016, http://dx.doi.org/10.1016/j.gca.2016.06.001). Comparisons are good except for Ce where their data are higher than ours.



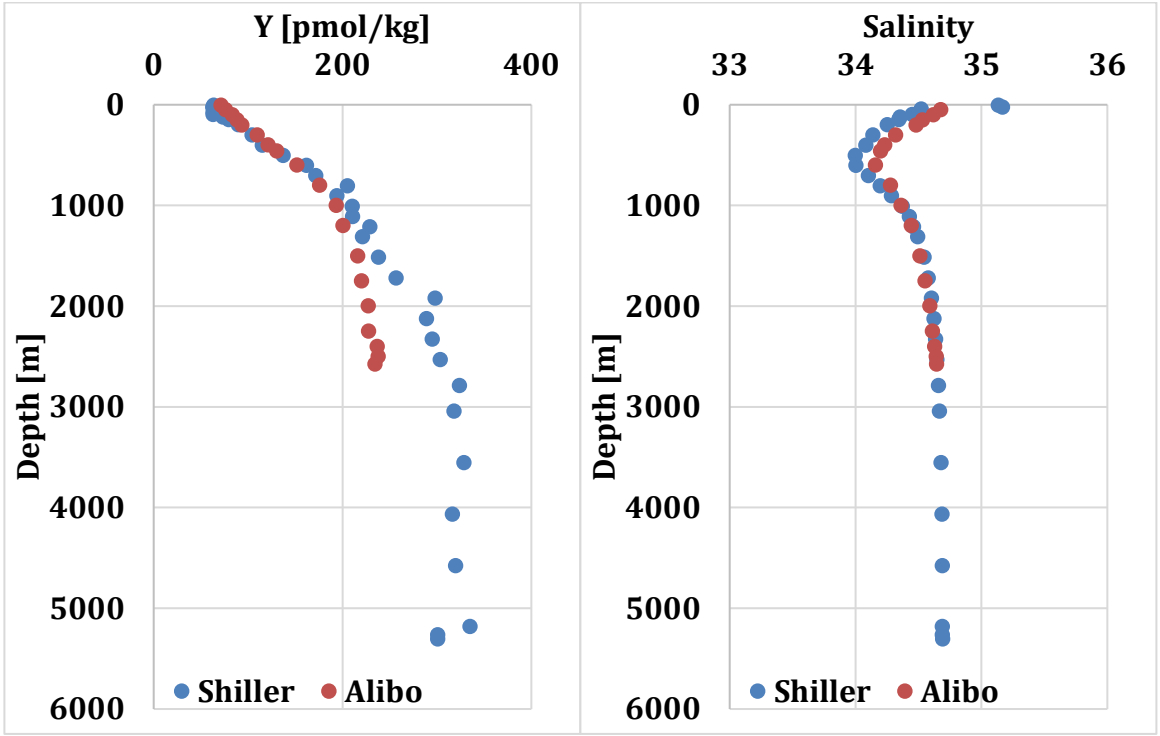
GP15-Station 35; 27N, 152W (Shiller) and VERTEX IV; 28N, 155W (DeBaar et al., Geochim. Cosmochim. Acta 2018, https://doi.org/10.1016/j.gca.2018.03.003)



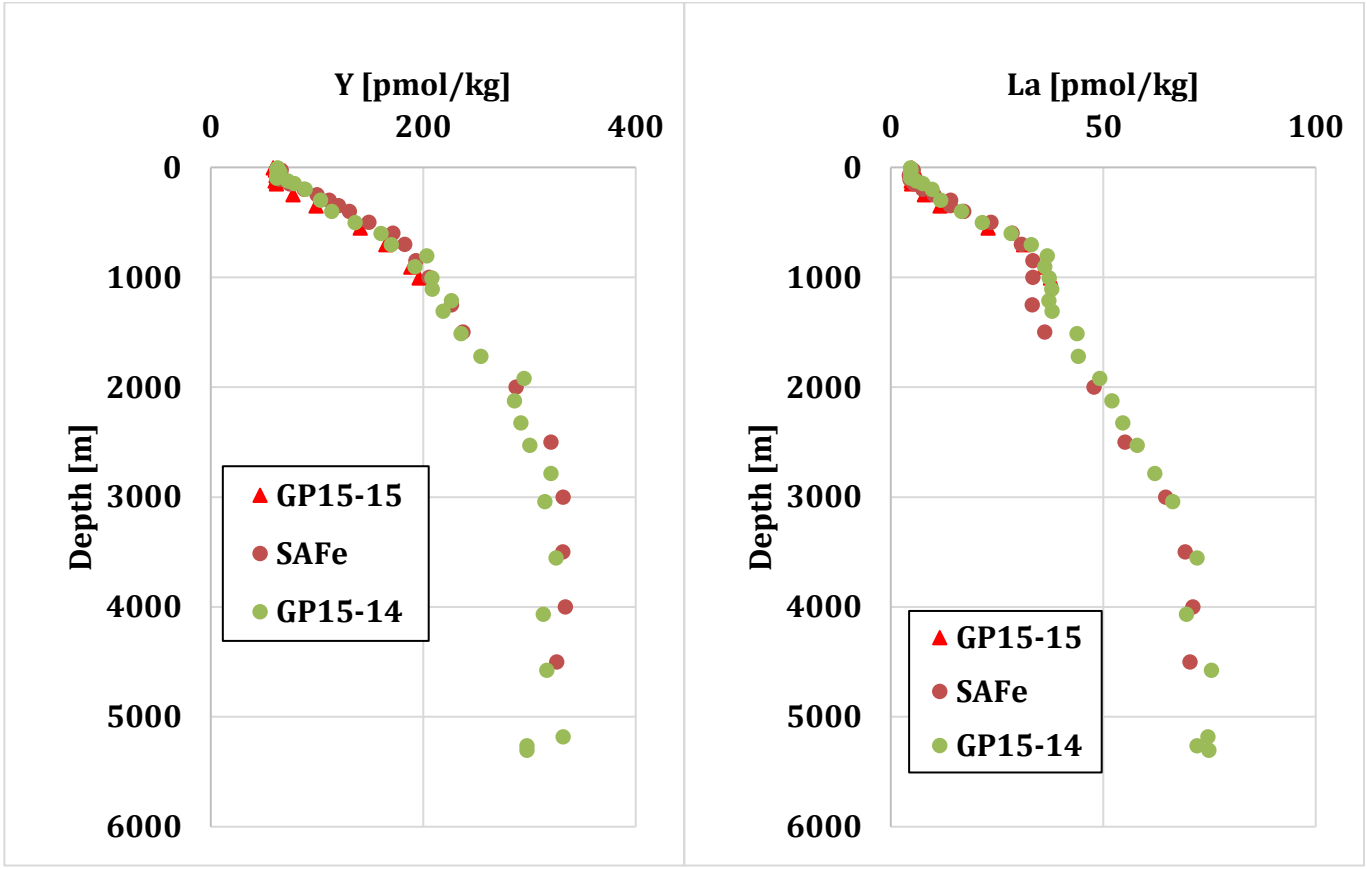
GP15-Station 5; 53.7N, 153.8W (Shiller) and CTD054-02; 54.3N, 155.5W (Haley et al., Geochim. Cosmochim. Acta 2014, http://dx.doi.org/10.1016/j.gca.2013.11.012). Comparisons are good except their deep water Ce is high and somewhat scattered compared to ours.



Comparisons for Yttrium: GP15 Station 14; 32N, 152W (Shiller) with Alibo-Nozaki (Geocim. Cosmochim. Acta 63 [1999] 363) 34N, 140E. These stations are at similar latitude but are far apart longitudinally (i.e, about 6000 km apart). Note that below ~1000 m the Alibo-Nozaki data for all REEs is lower than our GP15 data.

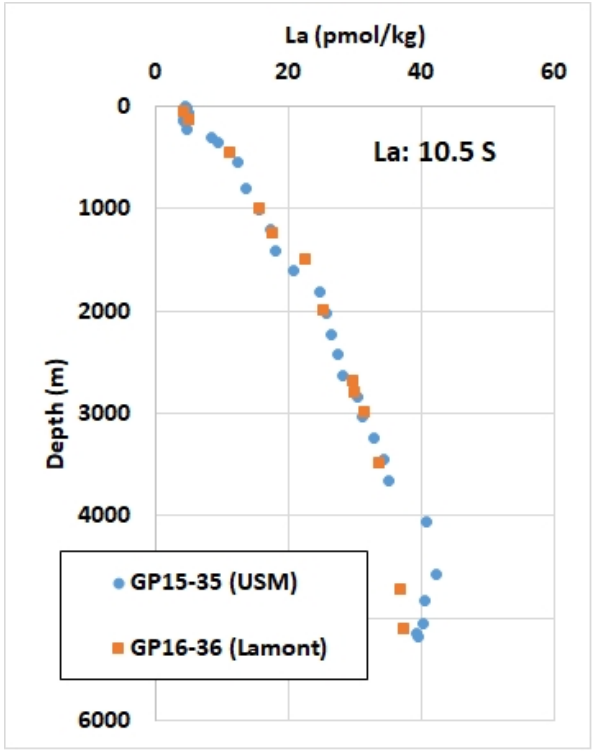


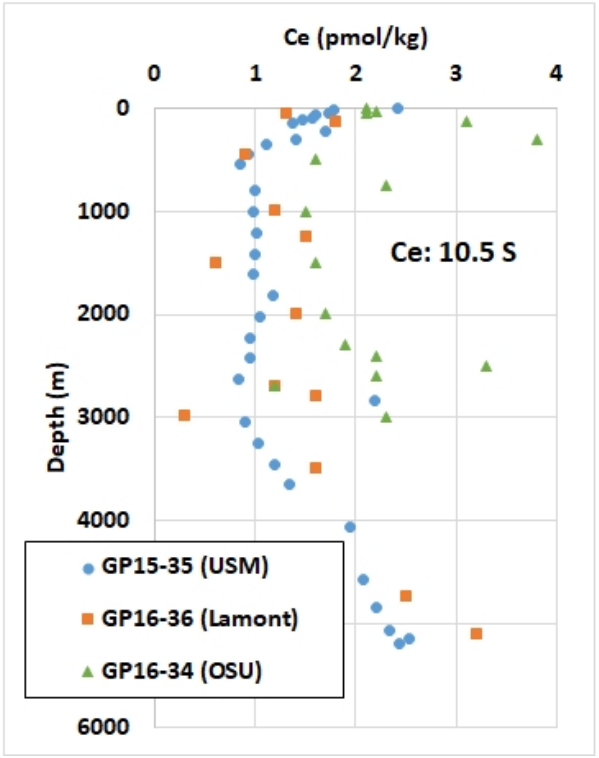
Comparisons for Y and La: GP15 Station 14 (32N, 152W) and Station 15 (29.5N, 152W) [Shiller, this work] with data from SAFe (30N, 140W) reported by Parker et al. (Geophys. Res. Lett. 2016, https://doi.org/10.1002/2016GL067827).

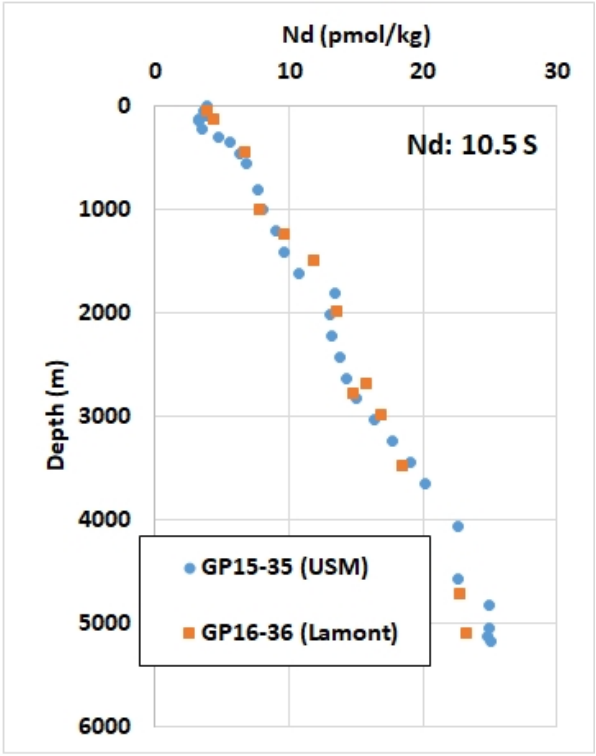


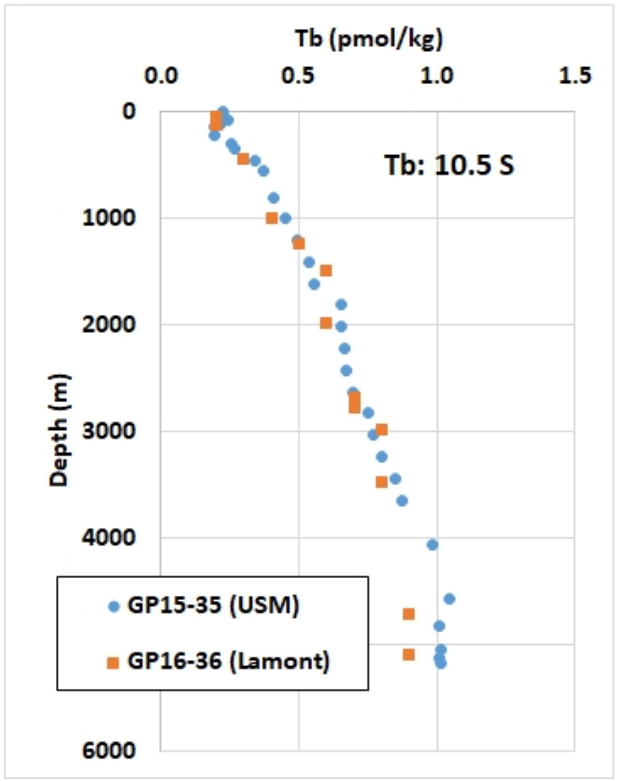
After the initial creation of this intercalibration report, data from the GP15-GP16 crossover station became available: Wu, Y., Basak, C., Pena , L. D., Bolge, L. L., Goldstein, S. L. (2023) Dissolved neodymium (Nd) isotopes and concentrations of rare-earth elements (REEs) reported by LDEO in seawater samples collected during the GEOTRACES GP16 East Pacific Zonal Transect cruise (TN303) on R/V Thomas G. Thompson from October to December 2013. Biological and Chemical Oceanography Data Management Office (BCO-DMO). (Version 1) Version Date 2023-12-19. doi:10.26008/1912/bco-dmo.915447.1

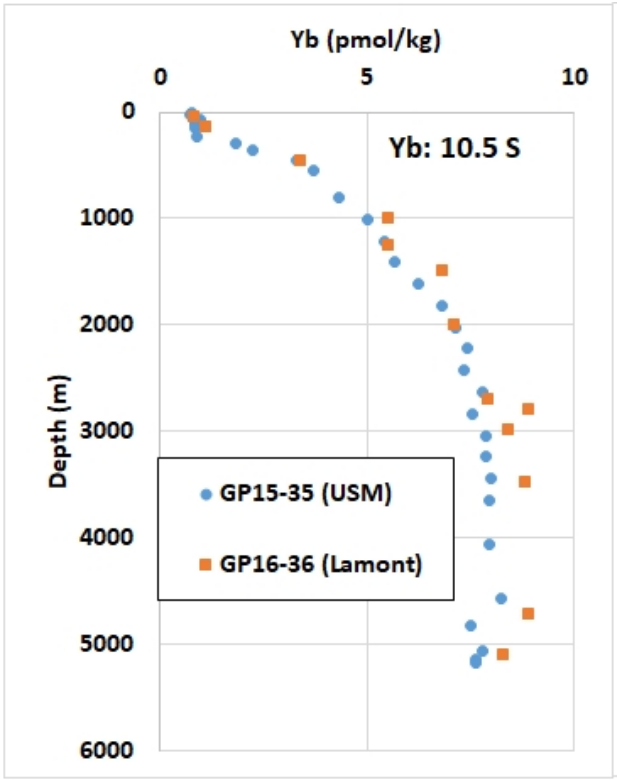
Below is the comparison for the 10.5 S, 152 W crossover at GP15 Station 35 (our data) and GP16 Station 36 Wu et al. (as cited above). Similar to the comparison (above) with the Haley et al. GP16 Station 34 data, there is generally good agreement. However, once again, the other dataset is more scattered for Ce and also for some of the heavier REEs. In fact, there is not great agreement between the Lamont (GP16 Stn 36 Ce data from Lamont and the GP16 Stn 34 Ce data from OSU). (See next page for Tb, Yb, and Lu.)

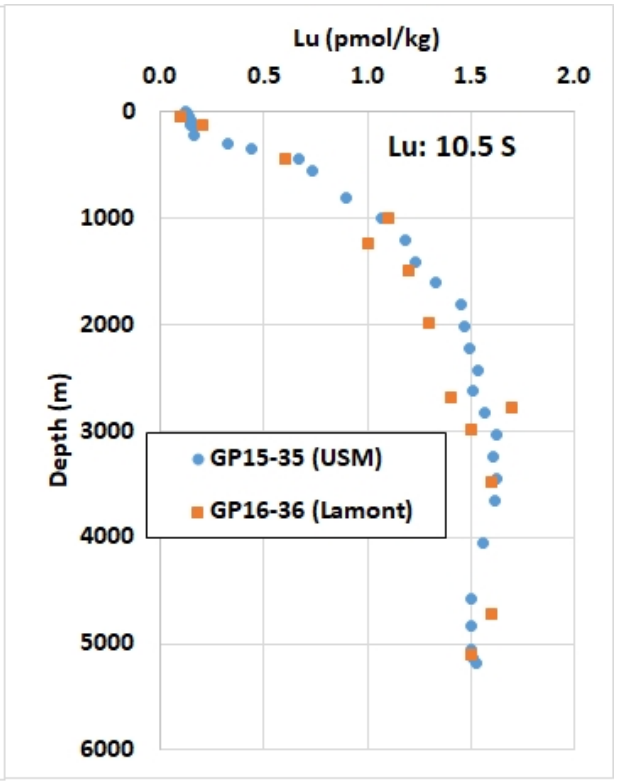







Tb (pmol/kg)
0.0
0.5
1.0
1.5
Tb: 10.5 S
Depth (m)
0
1000
2000
3000
4000
6000
GP15-35 (USM)
GP16-36 (Lamont)


Yb (pmol/kg)
0
5
10
Yb: 10.5 S
Depth (m)
0
1000
2000
3000
4000
5000
6000
GP15-35 (USM)
GP16-36 (Lamont)


Lu (pmol/kg)
0.0
0.5
1.0
1.5
2.0
Lu: 10.5 S
Depth (m)
0
1000
2000
3000
4000
5000
6000
GP15-35 (USM)
GP16-36 (Lamont)